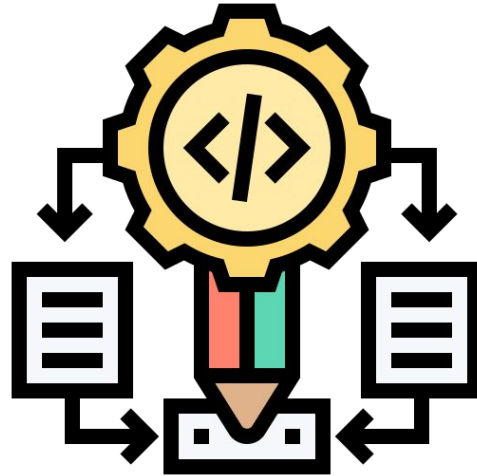


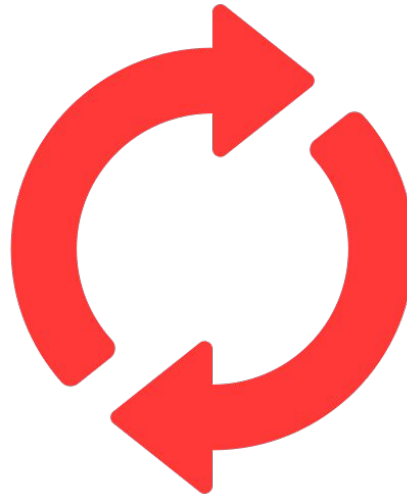
Algorithm 1



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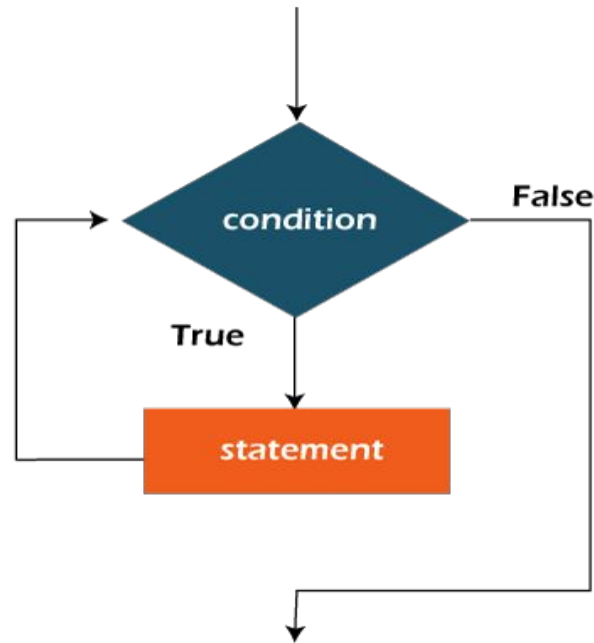
Loops

حلقات



Loops

Loops are used in programming to repeat a block of code multiple times



Loops

In most programming languages, there are mainly three types of loops :

1. for loop
2. while loop
3. do-while loop
- 4. Foreach loop**

Loops- For

The for loop is used to repeatedly execute a block of code a specified number of times

```
for (initialization; condition; increment/decrement) {  
  
    // code to be executed in each iteration  
  
}
```

Loops- For

Algorithm Syntax	C Syntax
<pre>Algorithm Example Begin for i := 1 to 10 do // Loop body End</pre>	<pre>#include <stdio.h> int main() { int i; for (i = 1; i <= 10; i++) { // Loop body } return 0; }</pre>

Loops - For

```
#include <stdio.h>
```

```
int main() {
```

```
    for (int i = 1; i <= 10; i++) {  
        // Print the current value of i  
        printf("%d\n", i);  
    }
```

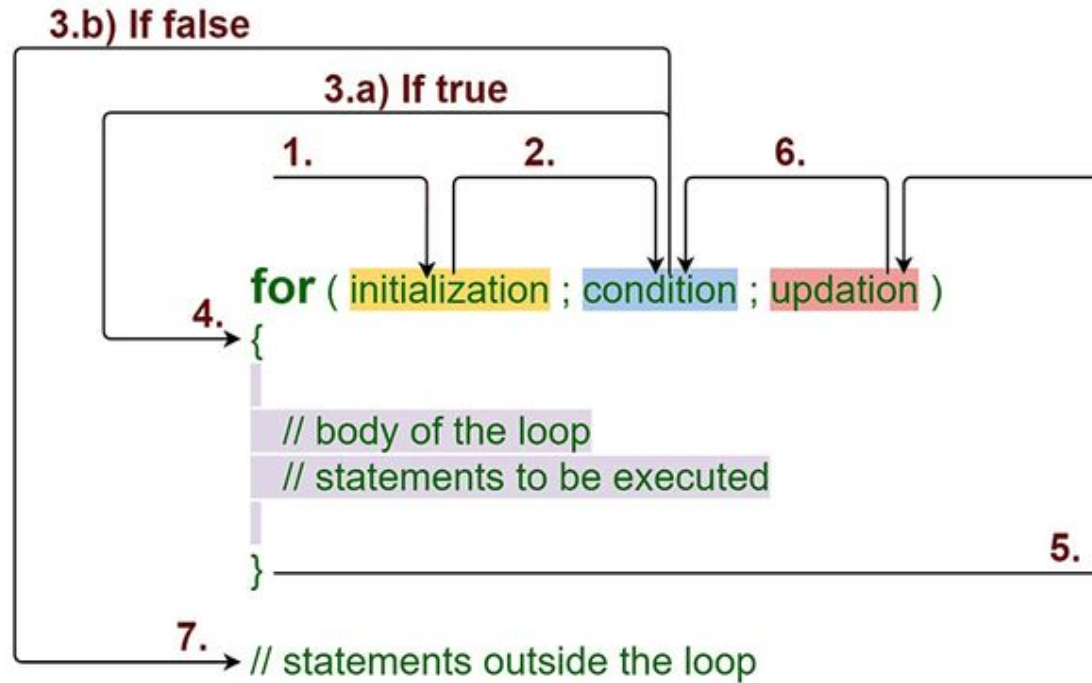
```
    return 0;
```

```
}
```



```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

For Loop



Loops- while

The while loop is used to repeatedly execute a block of code as long as a condition is true

```
while (condition) {  
  
    // code to be executed in each iteration  
  
}
```

Loops- while

Algorithm Syntax	C Syntax
<pre>Algorithm Example Begin while x < 10 do // Loop body End</pre>	<pre>#include <stdio.h> int main() { int x = 0; while (x < 10) { // Loop body x++; } return 0; }</pre>

Loops- while

```
#include <stdio.h>
void main() {
    // Initialize the counter variable
    int i = 1;

    // Continue looping as long as i is less than or equal to 10
    while (i <= 10) {
        // Print the current value of i
        printf("%d\n", i);
        // Increment i by 1 after each iteration
        i++;
    }
}
```

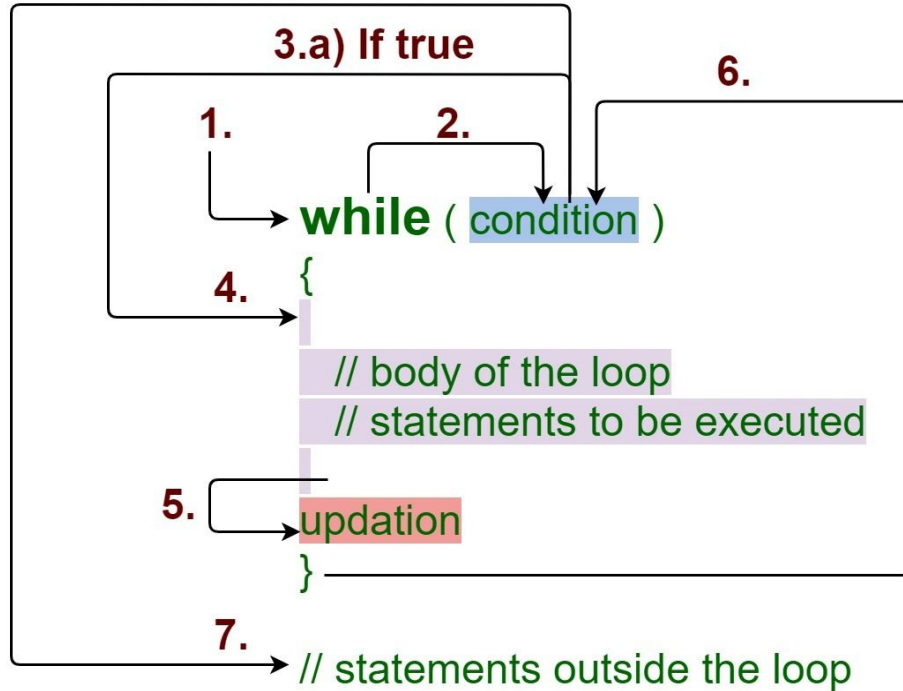


1
2
3
4
5
6
7
8
9
10

While Loop

3.b) If false

3.a) If true



Loops - do-while

The do-while loop is used to repeatedly execute a block of code at least once, and then continue execution as long as a condition is true.

```
do {
```

```
    // code to be executed in each iteration
```

```
} while (condition);
```

Loops- do-while

Algorithm Syntax	C Syntax
<pre>Algorithm Example Begin repeat // Loop body until x >= 10 End</pre>	<pre>#include <stdio.h> int main() { int x = 0; do { // Loop body x++; } while (x < 10); return 0; }</pre>

Loops- do-while

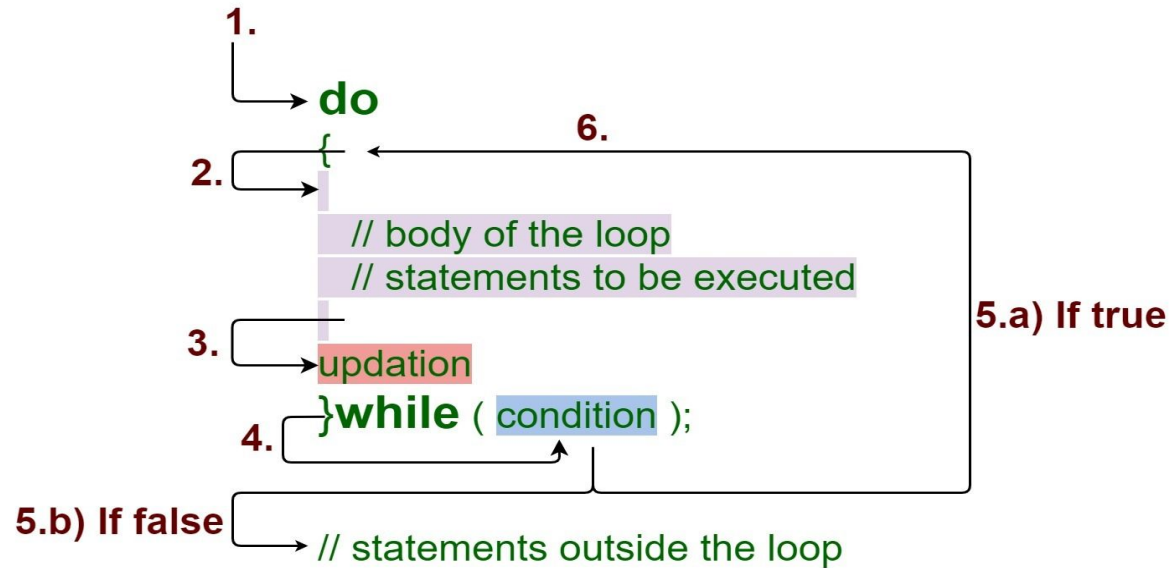
```
#include <stdio.h>
void main() {
    int i = 1; // Initialize the counter variable

    do {
        printf("%d\n", i); // Print the current value of i
        i++; // Increment i by 1
    } while (i <= 10);
}
```



1
2
3
4
5
6
7
8
9
10

Do - While Loop





Thank
You!

